

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MAR 311 — Past Questions & Materials
by Renance

Course: MAR 311
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 3
Levels: 300

This study pack was compiled and restructured by Renance for MAR 311 students at Federal University of Petroleum Resources, Effurun. It brings together 3 past paper(s)/material(s) into one clean, readable document so you can revise without chasing scattered PDFs. Practise these same questions interactively on Renance with instant answers, AI explanations, and full CBT mock exams — over 235,000+ questions across JAMB, Post-UTME, WAEC, NECO and university courses.

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Paper 1 of 3 — MAR 311 Intro to oceanography 2024/2025.

Level 300 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES EFFURUN DELTA STATE

DEPARTMENT OF MARINE ENGINEERING

MAR 311: INTRO. TO OCEANOGRAPHY: 300 LEVEL FIRST SEMESTER

2024/2025 ACADEMIC SESSION- MAY, 2025

TIME 2 HOURS. ANSWER 4 QUESTIONS. EXCEPT WHERE STATED, ALL QUESTIONS CARRY EQUAL MARK OF 4.

Question 1

- A. Discuss the "ocean ring of fire" of world ocean.
- 8B. What is the major explanation to the causes of deep trenches.
- C. Why is the middle of ocean not deepest as one will expect?
- D. Discuss any bio significant of volcanic eruption in the deep oceans
- E. Differentiate between sea and ocean. *

Question 2

- 4. Compare corrosion effects on structures at 100m and 10m of ocean depths.
- 5. Describe the earth without the ocean.
- C. What exactly will happen today if the sun ceases to exist?
- D. Why do we say that tides in west Africa is semi diurnal?
- E. The Gulf of Mexico and part of Malaysia and Indonesia have diurnal tides. What is diurnal tide? And what could be the possible reason(s) it happens compared to the semi diurnal system?

Question 3

- 4, With simple charts, describe the relationship between salinity and temperature, density and temperature.
- B. Usually, motion of fluid is from area of high pressure to areas of low pressure. For both wind and the ocean body, describe agents and processes that can produce wind and ocean water high pressures. 8mks
- C. Describe light penetration in the ocean and argue to demonstrate that the ocean is blue.
- ©. Why is your stream somewhat greenish?

Chatkes

Question 4

- A. Consider that you woke up to see your water tanker of very low freeboard heading to the Bermuda triangle. In-fact it is unavoidably in the triangle in the next an hour. What actions can you take to mitigate casualties and why? Freeboard is the distance between the sea water and the deck. 6mks

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B. In a single sentence state what caused the titanic verse! to sink? 3mks

C. The strongest underwater current that operates between MaGagascar ano southern Afric called? While the one that dominates west Africa is? 8mks_

D. Why Is the river Niger not blue. 3mks

Question 5

A. Name three causes gires?

B. Why Is ocean salty and rivers fresh water?

C, Differentiate between hurricane, tonaldoes and rogue waves.

D. What is the condition for slamming waves in terms of wave height and wave period?

E. With aid of diagram, show how wave and slamming forces differ.

Question 6

A. With the ald of a diagram, describe the concept of Easterlies, Westerlies and trade winds in the global wind pattern. 12mks

B. Explain the wet and dry seasons of Nigeria using the trade winds. Using a sketch will attract +2 marks. 4mks

C. The gulf stream plays a significant role in global weather pattern between the tropics and temperate world. How? 4mks

Goodluck!

Paper 2 of 3 — MAR 311 INTRO TO OCEANOGRAPHY 2023/2024 PQ

Level 300 • First Semester • PastQuestion

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| 2023/2024 ACADEMIC SESSION MAY, 2024 x OF SMARKS

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a. m queedioa Id, what could be (6 consequences fur he maritun' sector?

—B.Trudy, we say that offshore wes) Africa iS "kind"

c, List two Causes of ocean pyres.

d. List the World ocean amd pyres a5 discussed in the

» Discuss

Clugs-

4,

a Where ate the trenches located? list 2 important characteristics Of the trenches. |

b, with aid ur 3 drawing, differentiate between typical sea bottom of the atlantic and pacific oceans

€. Why #6 ther@ ice and snow in South Africa but none Gham?

G, Discuss (he variabilities in the salinity of the world Oçean

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gee 8 SOR havt heard very awful stories about the bem vgle. istherea suggestion you can off ef to solve the

mystery? |

», Oisou') the Globbs! winds of the world and local winds in| 2çria. Please use drawings.

—& Nhat (fects do these winds have on the world and local cllmates?

'go 6. Whe! s the ditierence between tidal currant and i drifting current.

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4. YOu. .°@ involved in the tidal measurements in Agbarho riyur. Discuss the results of the work, Remember to mention

the ave 'ege tidal range in your discussions.

oO. t hezc a professor from Uniport argue that deep ocean wa: ~sgreen. Please agree or disagree with reasons

¢. Lise. ss. how deep visible figh. can travel in water,

G, Oiscvss the importance of preserving and caring for the ocean as a direct function of man's existence on Earth

3 Whei are the Causes of monster waves? |

b. Cornpere fegular waves with irregular we-«s of the ocean.

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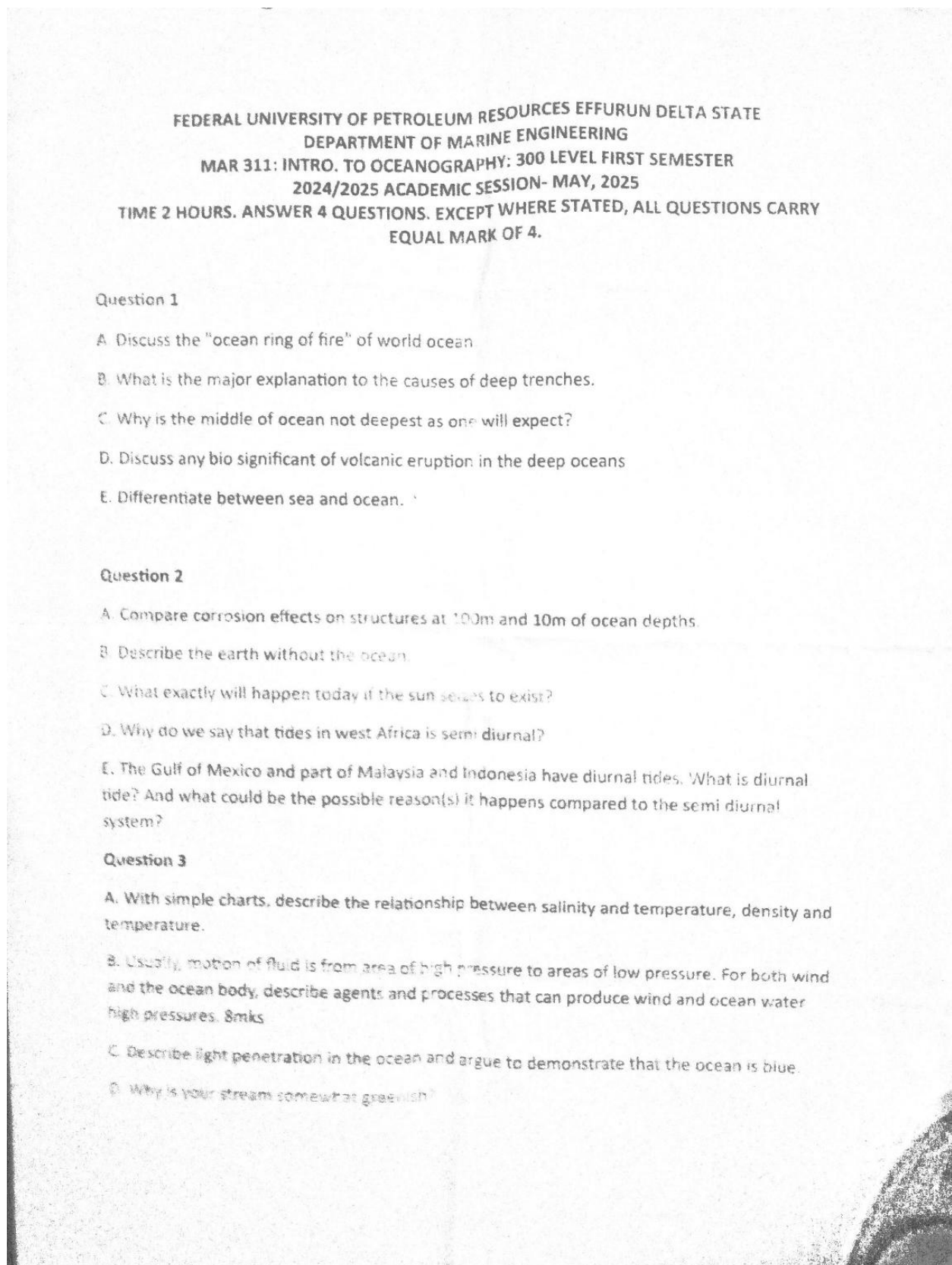
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Paper 3 of 3 — MAR 311 INTO. TO OCEANOGRAPHY 2025

Level 300 • First Semester • PastQuestion

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